

PART 4: SEQUENTIAL LOGIC

ANALYSIS OF SEQUENTIAL CIRCUITS

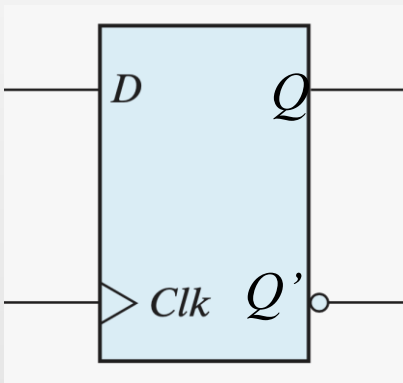
SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

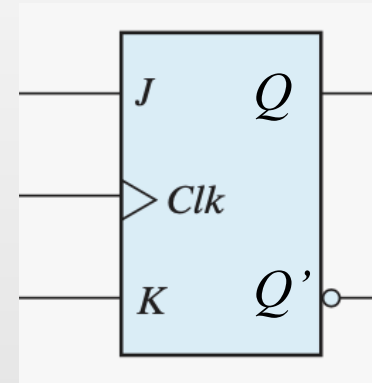
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FLIP FLOPS



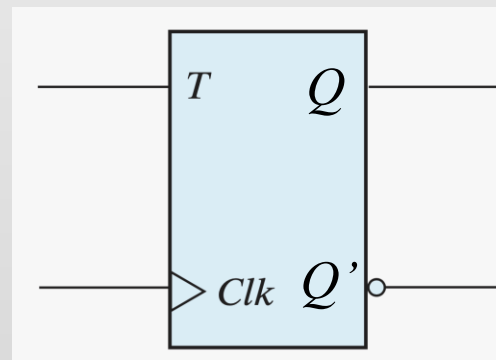
D	Q(t + 1)	
0	0	Reset
1	1	Set

$$Q(t + 1) = D$$



J	K	Q(t + 1)	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

$$Q(t + 1) = JQ' + K'Q$$



T	Q(t + 1)	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

$$Q(t + 1) = TQ' + T'Q$$



ANALYSIS OF CLOCKED SEQUENTIAL CIRCUITS

- Analysis describes what a given circuit will do under certain operating conditions.
- The behavior of a clocked sequential circuit is determined from the inputs, the outputs, and the **state** of its flip-flops.
- The outputs and the next state are both a function of the inputs and the present state.
- The analysis of a sequential circuit consists of obtaining a table or a diagram for the time sequence of inputs, outputs, and internal states.



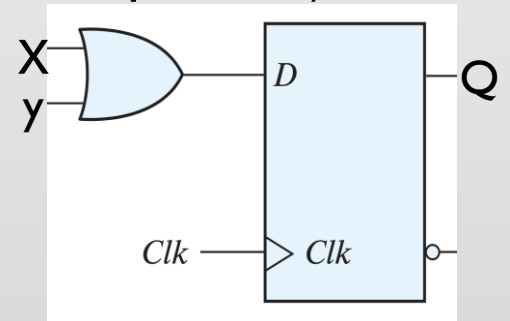
ANALYSIS OF CLOCKED SEQUENTIAL CIRCUITS

- Analysis of sequential circuits requires either:
 - Obtaining a table
 - Timing diagram of inputs, outputs and internal states
 - Boolean expression (with time sequence)
- How to recognize a clocked sequential circuit logic diagram?
 - Any types of flip-flops with clocked inputs. May or may not include combinational logic gates.

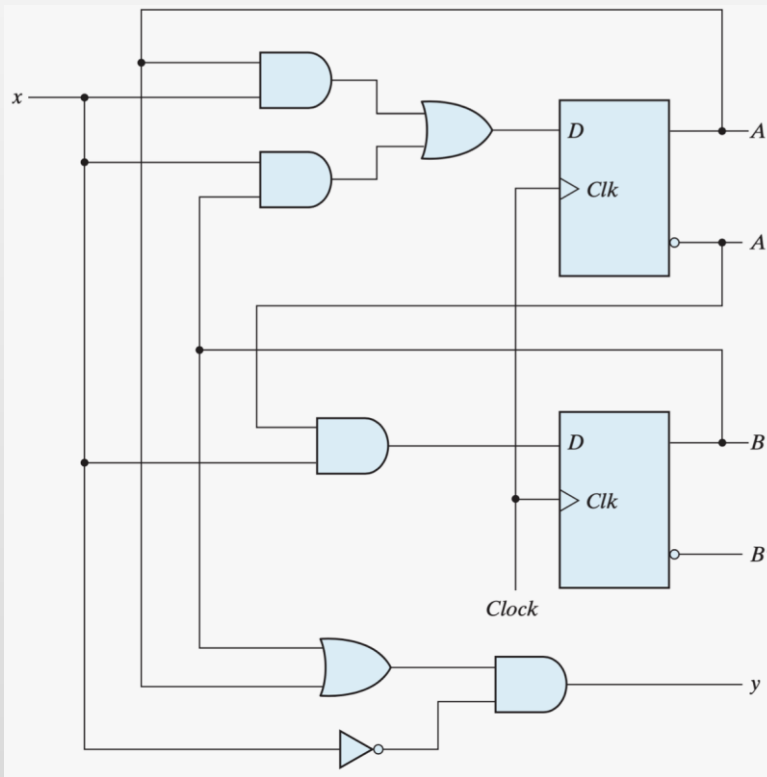
FLIP FLOP INPUT EQUATIONS

- The logic diagram of a sequential circuit consists of flip-flops and gates.
 - The interconnections among the gates form a combinational circuit.
 - Combinational circuit region which generates external outputs are described by Boolean functions
→ Output Equations
- The circuit which generates the inputs to FFs → Input Equations (Excitation equations)
- Convention:
 - Input FF symbol → input equation variable
 - Subscript → FF output

$$D_Q = x + y$$



FLIP FLOP INPUT EQUATIONS



- Write the FF input equation and the output equations of the circuit.
- Convention:
 - Input FF symbol $\rightarrow$ input equation variable
 - Subscript $\rightarrow$ FF output

$$D_A = Ax + Bx$$

$$D_B = A'x$$

$$y = (A + B)x'$$

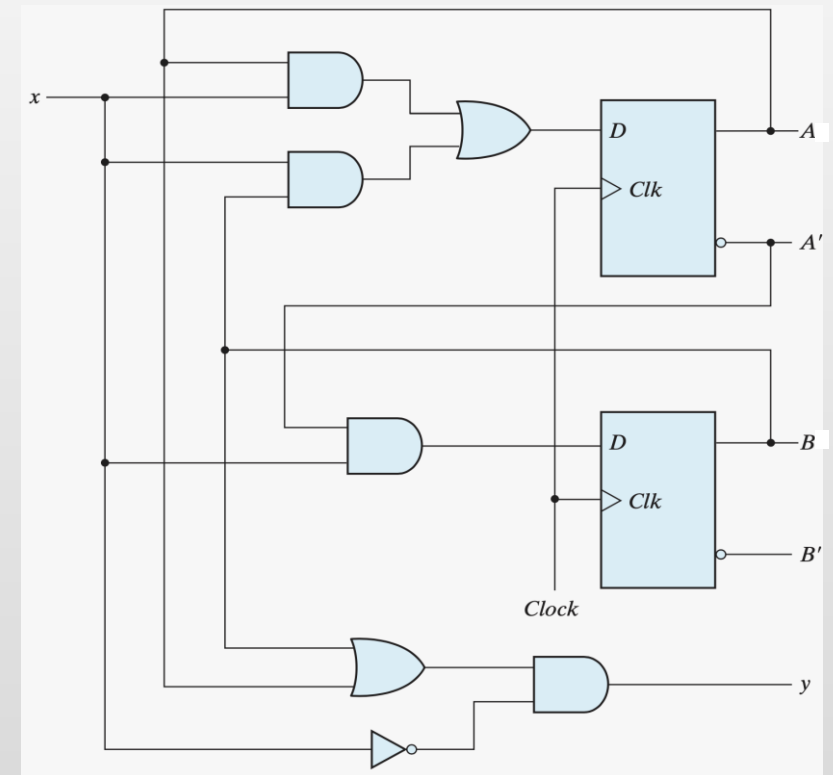


STATE EQUATIONS

- The behavior of a clocked sequential circuit can be described algebraically by means of **state equations**.
- A **state equation** (also called a **transition equation**):
 - Specifies the next state as a function of the present state and inputs.
 - Is an algebraic expression that specifies the condition for a flip-flop state transition.
 - **Example:** $y(t+1) = y(t) + z(t)$
- The left side of the equation, with $(t + 1)$, denotes the next state of the flip-flop (y) one clock-edge later.
- The right side of the equation is a Boolean expression that specifies the present state and input conditions that make the next state equal to 1.

STATE EQUATIONS

- $A(t + 1) = A(t)x(t) + B(t)x(t)$
- $B(t + 1) = A'(t)x(t)$
- Since all the variable are a function of the present state, designation (t) can be ignored.
- $A(t + 1) = Ax + Bx$
- $B(t + 1) = A'x$
- $y(t) = [A(t) + B(t)]x'(t)$
- $y(t) = (A + B)x'$



STATE EQUATIONS

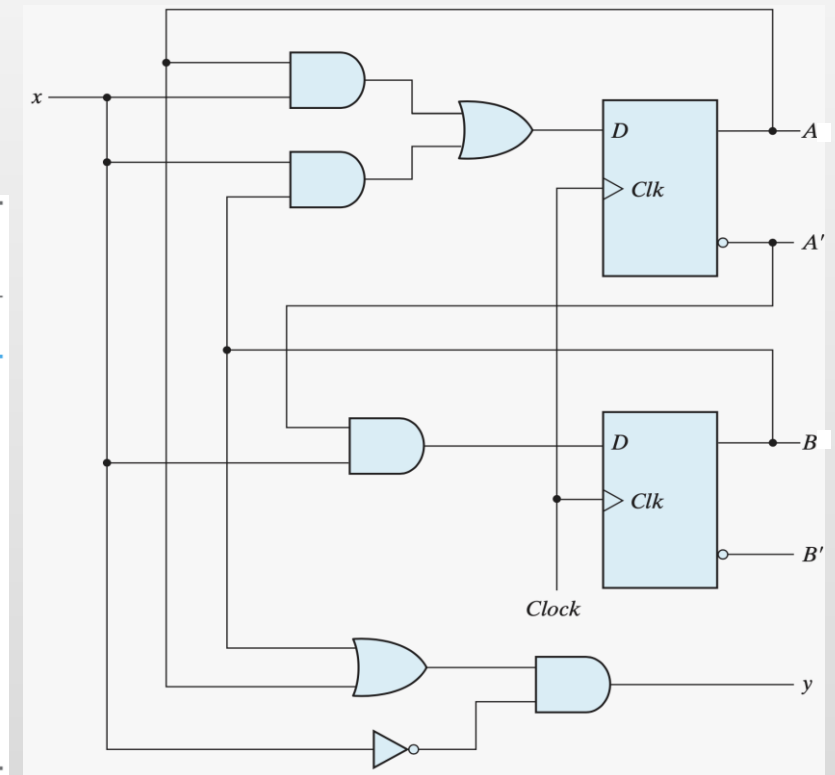
- The time sequence of inputs, outputs, and flip-flop states can be enumerated in a *state table* (sometimes called a *transition table*).
- Requires all possible combinations of present states and inputs.

$$A(t + 1) = A(t)x(t) + B(t)x(t)$$

$$B(t + 1) = A'x$$

$$y(t) = (A + B)x'$$

Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0





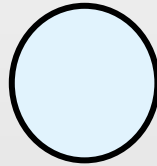
STATE TABLE

- A sequential circuit with m flip-flops and n inputs needs 2^{m+n} rows in the state table.
- The binary numbers from 0 through $2^{m+n} - 1$ are listed under the present-state and input columns.
- The next-state section has m columns, one for each flip-flop.
- The binary values for the *next state* are derived directly from the *state equations*.
- The output section has as many columns as there are output variables.
 - It's binary value is derived from the circuit or from the Boolean function in the same manner as in a truth table.

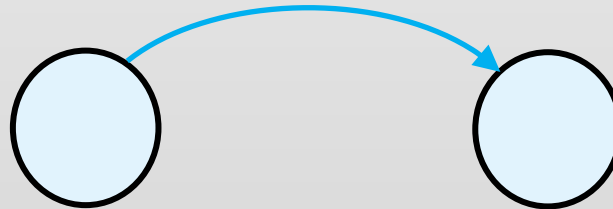
STATE DIAGRAM

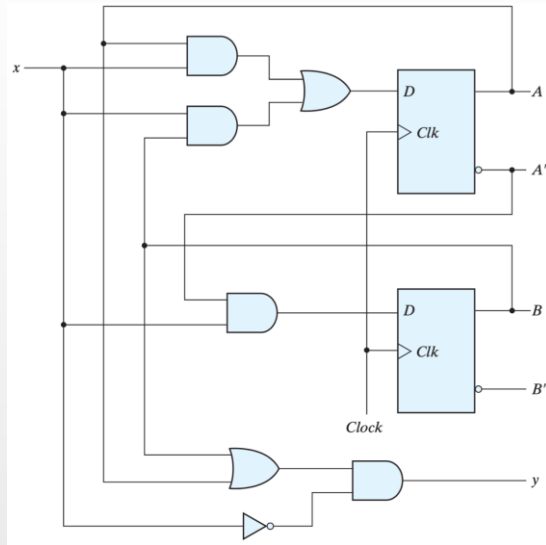
- State Table → State Diagram.

- State is represented by a circle.



- Transitions (Clock-triggered) between states are indicated by lines connecting the circles.



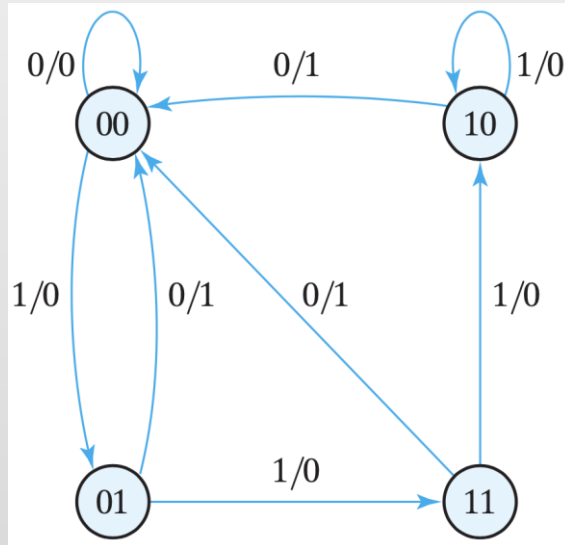


$$A(t + 1) = Ax + Bx$$

$$B(t + 1) = A'x$$

$$y(t) = (A + B)x'$$

Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0



Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0



ACTIVITY

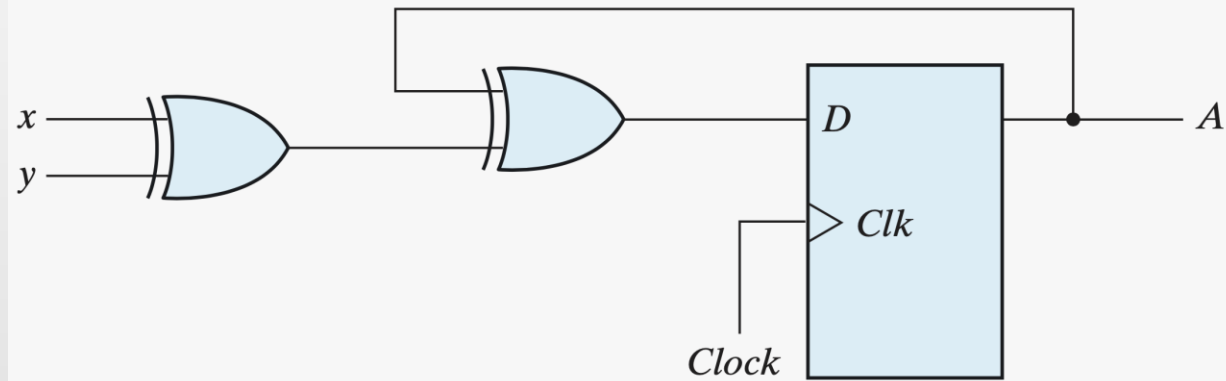
- Draw the (a) logic/circuit diagram, (b) state table and (c) state diagram for a clocked sequential circuit described by the following equation:

$$D_A = A \oplus x \oplus y$$

- Hints:
 - Inputs: x, y
 - Output: No equation given $\rightarrow$ Output comes from the FF

ACTIVITY

$$D_A = A \oplus x \oplus y$$



Present state	Inputs		Next state
<i>A</i>	<i>x</i>	<i>y</i>	<i>A</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



FLIP FLOP INPUT EQUATIONS – JK AND T

- For a D-type flip-flop, the state equation is the same as the input equation.
- When a JK or T flip-flop is used, it is necessary to refer to the corresponding characteristic table or equation to obtain the next state values.
 - JK equation $\rightarrow D = JQ' + K'Q$ T output $\rightarrow D = T'Q + TQ'$
- The next-state values can be derived as follows:
 1. Determine the flip-flop input equations in terms of the present state and input variables.
 2. List the binary values of each input equation.
 3. Use the corresponding flip-flop characteristic table to determine the next-state values in the state table.

FLIP FLOP INPUT EQUATIONS – JK

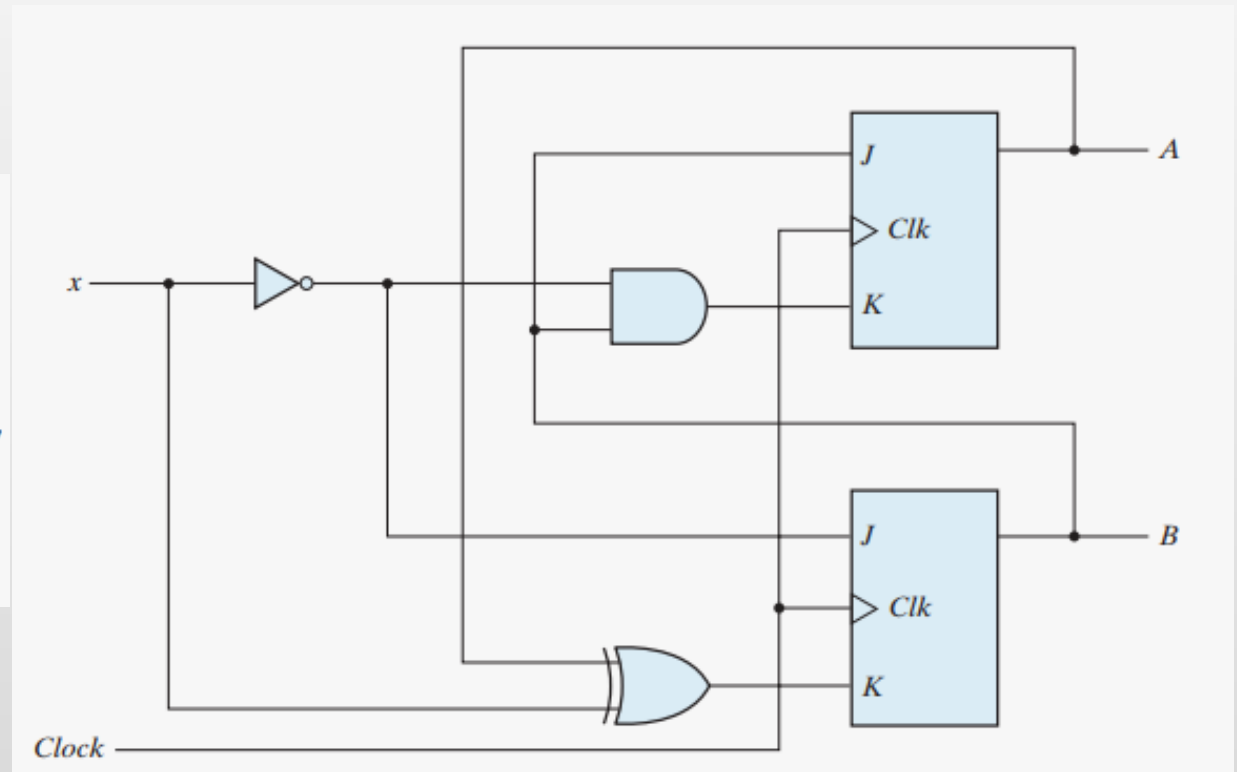
JK equation $\rightarrow D = JQ' + K'Q$

$$J_A = B \quad K_A = Bx'$$

$$J_B = x' \quad K_B = A'x + Ax' = A \oplus x$$

$$B(t+1) = x'B' + (A \oplus x)'B = B'x' + ABx + A'Bx'$$

$$A(t+1) = BA' + (Bx')'A = A'B + AB' + Ax$$





FLIP FLOP INPUT EQUATIONS – JK

$$J_A = B \quad K_A = Bx'$$

$$J_B = x' \quad K_B = A'x + Ax' = A \oplus x$$

$$B(t + 1) = x'B' + (A \oplus x)'B = B'x' + ABx + A'Bx'$$

$$A(t + 1) = BA' + (Bx')'A = A'B + AB' + Ax$$

Present State		Input	Next State		Flip-Flop Inputs			
A	B		A	B	J _A	K _A	J _B	K _B
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	1	1	1	1	0
0	1	1	1	0	1	0	0	1
1	0	0	1	1	0	0	1	1
1	0	1	1	0	0	0	0	0
1	1	0	0	0	1	1	1	1
1	1	1	1	1	1	0	0	0

FLIP FLOP INPUT EQUATIONS – T

■ What function does this circuit perform?

1. Flip flop input equations: $T_A = Bx$

$$T_B = x$$

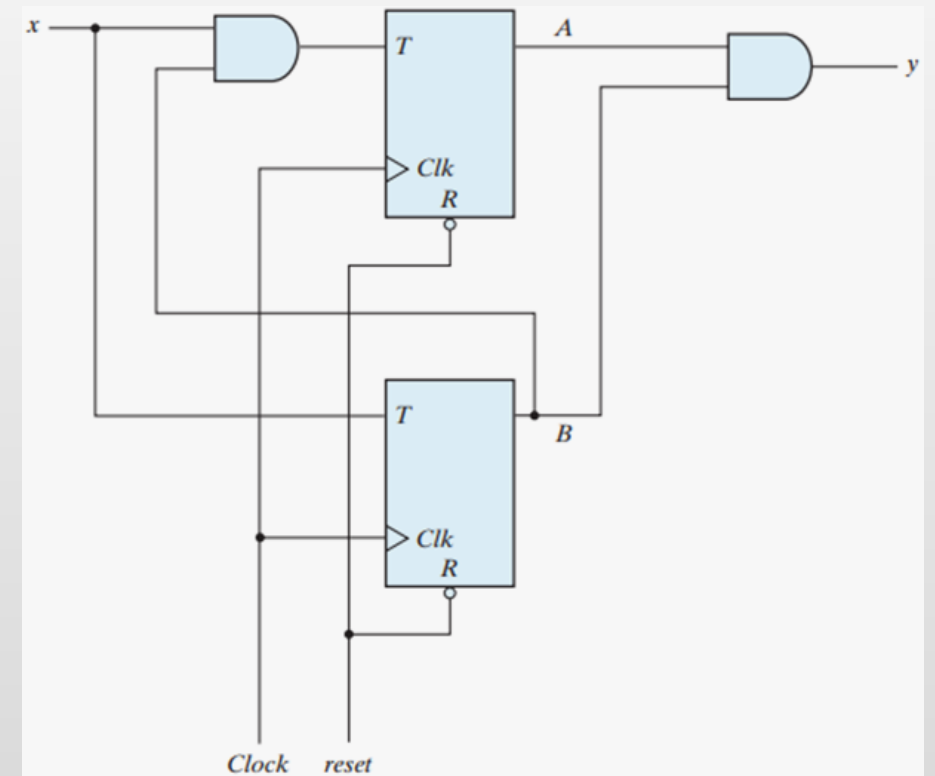
$$y = AB$$

2. State equations:

■ T output $\rightarrow D = T'Q + TQ'$

$$A(t + 1) = (Bx)'A + (Bx)A' = AB' + Ax' + A'Bx$$

$$B(t + 1) = x \oplus B$$



FLIP FLOP INPUT EQUATIONS –T

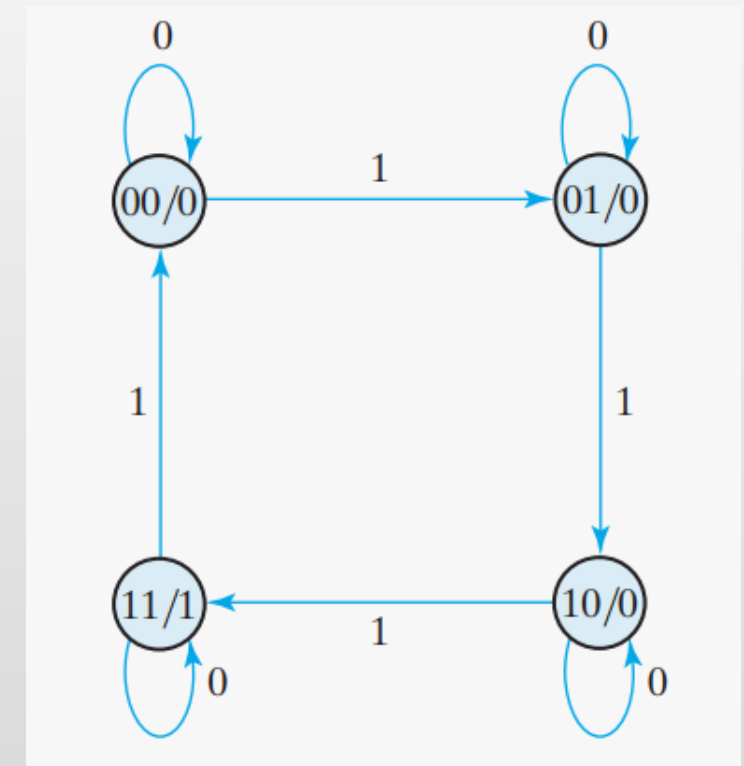
3. State table:

$$A(t + 1) = (Bx)'A + (Bx)A' = AB' + Ax' + A'Bx$$

$$B(t + 1) = x \oplus B$$

Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

State Diagram:



FLIP FLOP INPUT EQUATIONS

1. Determine the flip-flop *input equations* in terms of the present state and input variables.
2. Substitute the input equations into the flip-flop characteristic equation to obtain the state equations.
3. Use the corresponding state equations to determine the next-state values in the state table.
4. Draw a state diagram.

